

Anvil — Out-of-Sample Evidence Pack

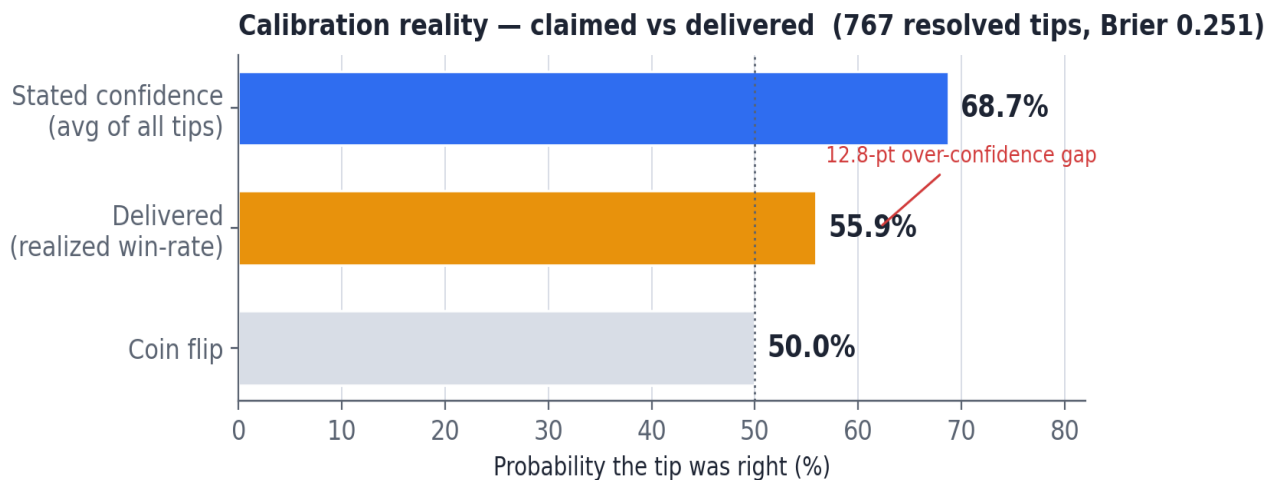
How the engine is tested, and what the test data says today. Walk-forward backtest on real NSE end-of-day option & equity history (Sep–Nov 2025).

984 tips issued in the walk-forward	767 resolved at expiry on realized prices	59 strategy cells measured	0 edge-verified (by design, see below)
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What this pack is

Anvil does not promise high directional accuracy — it promises **calibrated, audited probabilities**: when it says 70%, the event should happen about 70% of the time, shown on a live public reliability curve. This pack is the out-of-sample evidence behind that claim. For each past trading day the engine builds its signal context (look-ahead guarded), issues option and single-stock tips, then resolves each one on the **realized** settlement price held to expiry, **net of modelled round-trip cost** — so a tip can never win on paper alone.

The honest headline: calibration, not a hit-rate



Across all 767 resolved tips the engine stated an average **68.7%** chance of being right and was right **55.9%** of the time (Brier 0.251, where 0.25 = a flat 50% guess). On three months it is currently **over-confident and unproven** — which is exactly what the product says about itself rather than hides.

How it is tested (your “train / test / validation”)

Most Anvil signals are rule-based quant (Black-76 Greeks, cross-sectional momentum, HAR-RV), not a fitted model — so there is no weights “training” step yet (the ML meta-layer is a planned future wave). The equivalent, and **stricter**, evaluation is a walk-forward out-of-sample backtest with Combinatorial Purged Cross-Validation (CPCV) and an embargo, plus a multiple-testing-aware certification gate. Every reported number below is out-of-sample.

Gate criterion	Threshold	Why it exists
Sample size	$n \geq 50$ resolved / cell	enough independent trades to mean anything
Calibration	win-rate $\geq$ stated conviction	confidence must be honest, not inflated
Post-cost edge	> 0	profitable after realistic costs
Harvey t-stat	$ t \geq 3.0$	not a fluke; the bar mined-factor research demands

Deflated Sharpe	≥ 0.95	survives discount for number of strategies tried
Prob. of overfit (PBO)	≤ 0.5	the in-sample winner is not just overfit
Bootstrap 5th-pct edge	> 0	edge holds in a robust tail resample

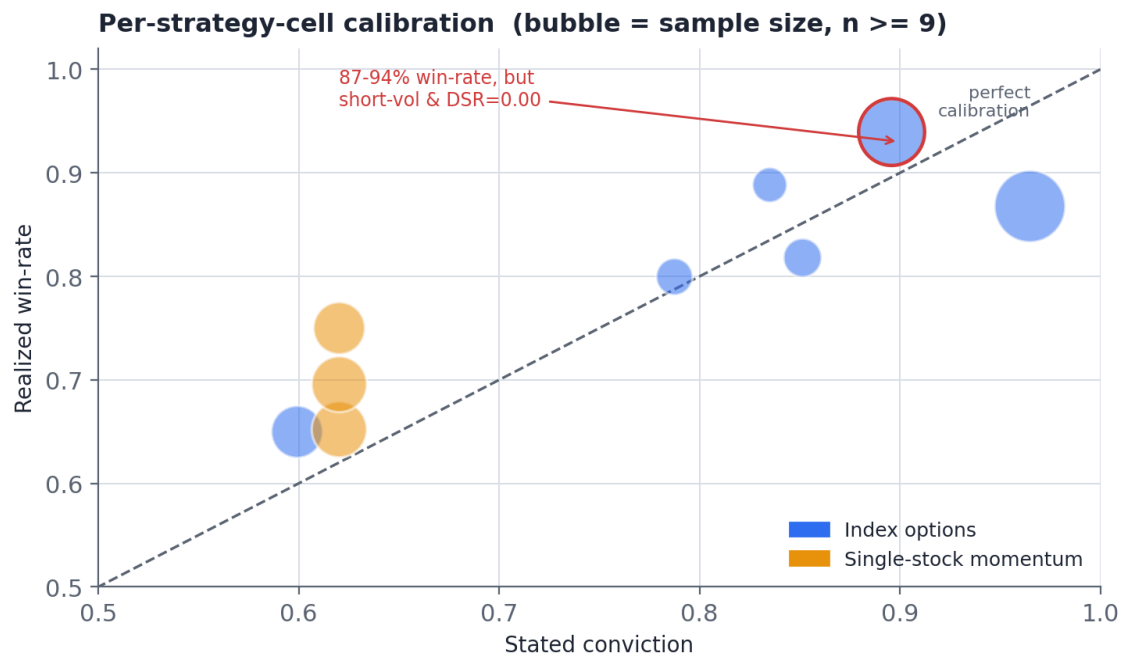
A cell is “edge-verified” only if it clears **every** row at once.

Track record by strategy cell (most-populated, $n \geq 9$)

Market	Strategy / regime	n	Win%	Conv %	Edge*	t	DSR	PBO	Verdict
NIFTY	short-strangle / event-crush	33	93.9	89.6	+0.25	2.33	0.00	0.49	Not verified
NIFTY	short-strangle / pin-low-vol	38	86.8	96.5	+0.24	3.48	0.00	0.49	Not verified
NIFTY	short-strangle / trend-high-vol	10	80.0	78.7	+0.19	1.64	0.00	0.49	Not verified
NIFTY	iron-condor / event-crush	11	81.8	85.1	+0.01	0.05	0.00	0.49	Not verified
NIFTY	directional-future / pin	20	65.0	59.9	+0.10	0.33	0.00	0.49	Not verified
BANKNIFTY	directional-future / trend	9	88.9	83.5	+2.57	3.50	0.00	0.49	Not verified
SBIN	equity momentum	23	65.2	62.0	+0.20	3.15	0.00	0.90	Not verified
MCX	equity momentum	23	69.6	62.0	+0.24	1.01	0.00	0.90	Not verified
HAL	equity momentum	20	75.0	62.0	+0.16	1.40	0.00	0.90	Not verified

*Edge = average post-cost return on risk per trade (e.g. +2.57 = +257% on capital-at-risk, typical of a leveraged directional future). DSR = Deflated Sharpe (needs ≥ 0.95); PBO = probability of backtest overfitting (needs ≤ 0.5). Every cell fails at least one test — the gate is doing its job on three months of data.

Why the 87–94% win-rates are not the headline



Read this before quoting any win-rate to an investor

The 87–94% cells are **short-volatility** structures (selling strangles / condors): they win small very often and lose big rarely — “picking up pennies in front of a steamroller.” A high win-rate is **not** a profitable edge, which is precisely why every one shows Deflated-Sharpe 0.00 and stays **not verified**. There is also no “800% accuracy” metric — win-rate caps at 100%; 800% could only be the *return* on a single leveraged winning trade, and one trade is not evidence.

Specimen resolved trades (illustrative, not the evidence)

Issued	Market	Spot at call	Resolved	Stated p(win)	Outcome
2025-09-01	NIFTY	24,625	2025-09-09	78%	WIN
2025-09-01	BANKNIFTY	54,003	2025-10-28	99%	WIN
2025-09-01	BANKNIFTY	54,003	2025-09-30	65%	LOSS
2025-09-01	NIFTY	24,625	2025-09-30	65%	LOSS
2025-09-02	NIFTY	24,580	2025-09-16	60%	WIN

The path to a verified edge

Three months (~62 trading days, Sep–Nov 2025) is too little: option tips resolve in same-day clusters, so the count of **independent** days is small and the Deflated-Sharpe / PBO battery cannot certify anything. The unlock is more history, not a new model: fetch multi-year NSE F&O bhavcopy plus index/stock closes, re-run the same walk-forward, and any cell whose edge is real will accumulate the independent sample needed to clear the gate — at which point it earns the “Edge-verified” badge automatically. The companion runbook gives the exact commands.

The investor takeaway

Anvil's differentiator is not a bigger accuracy number — it is the only honest one: a rigorous anti-overfitting gate, a live public reliability curve, and abstention when there is no edge. That survives technical diligence; a “70–800% accuracy” claim on an options tool does not.

Source: walk-forward backtest, source-class `tip_backtest` (real EOD only; synthetic/demo/paper data is firewalled out). Data: `anvil_store.duckdb` (`tip_validation`) & `anvil_ledger.duckdb` (forecasts/outcomes). Methodology: `anvil/backtest/validation.py`, `aggregate.py`, `tip_backtest.py`. No order execution; no investment advice.